

Feedback on two methodological choices (use/mention split, LLM annotation validation) and a suggestion on which venue to target.

The problem

A fine-tuned Italian BERT (UmBERTo, 125M params) reaches 99.75% accuracy in-domain on email fraud, but collapses to 2.43% on held-out SMS under leave-one-channel-out (LOO) validation — a 41-fold degradation. The model learned the style of the training channel, not the semantics of fraud. We call this the channel shortcut. CIC-Intents improves the LOO SMS number to 0.243, still low but far above chance.

Four Italian channels: 4,376 preprocessed → 950 annotated for IFIT

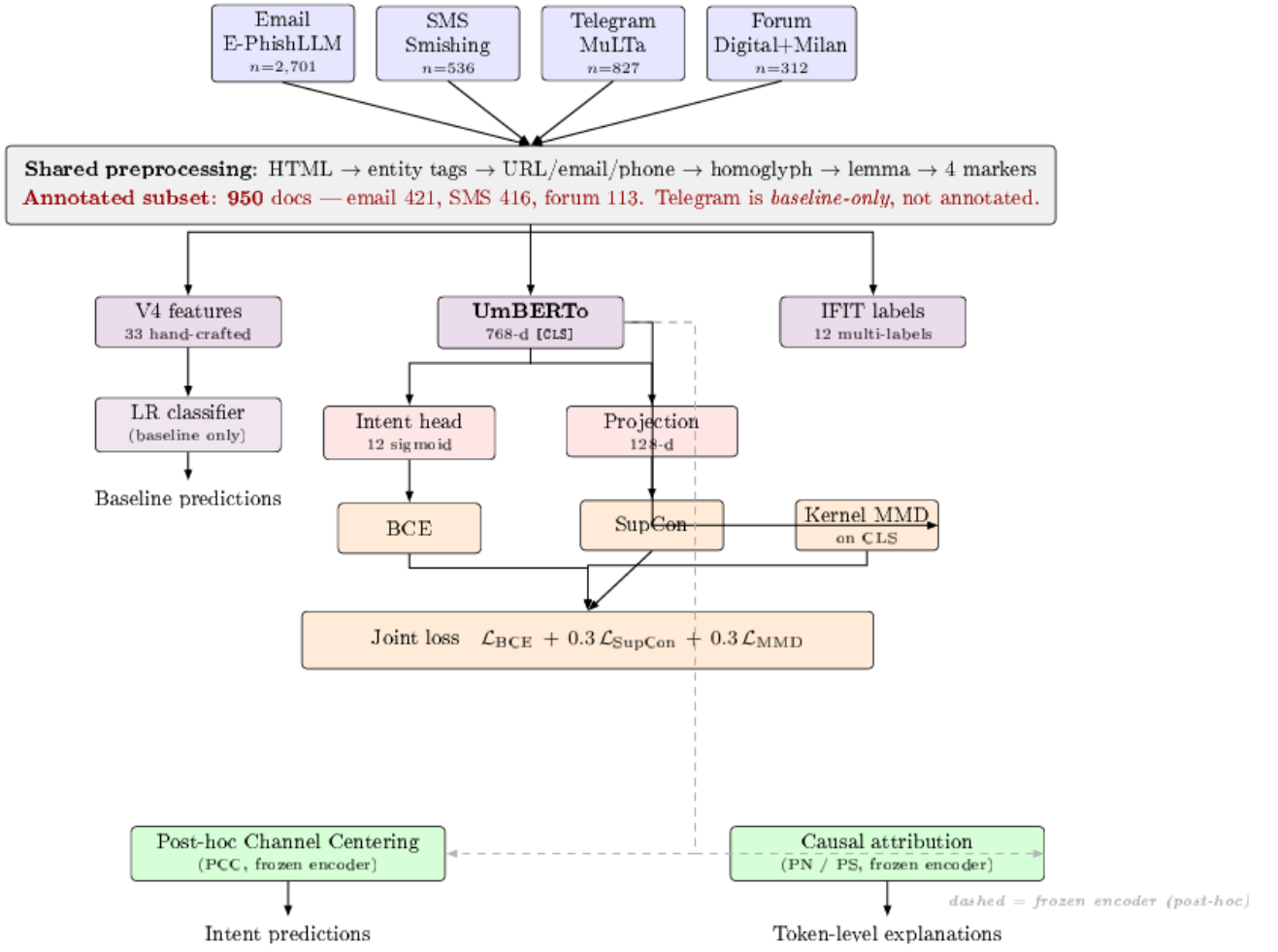
| Channel              | Preprocessed | Annotated | Role             |
|----------------------|--------------|-----------|------------------|
| Email (E-PhishLLM)   | 2,701        | 421       | use / legit      |
| SMS (Smishing-IMC25) | 536          | 416       | use              |
| Forum                | 312          | 113       | mention          |
| Telegram (MuLTa)     | 827          | 0         | legit proxy only |
| Total                | 4,376        | 950       |                  |

Why the drop from 4,376 to 950. Telegram (827) is retained only as a cross-channel legit proxy and carries no IFIT labels — it is never part of training or intent evaluation. Email drops from 2,701 to 421 because E-PhishLLM is LLM-generated: the same prompt (Intesa Sanpaolo, Poste Italiane, UniCredit...) produces hundreds of near-identical messages, and deduplication on `text_clean` leaves only unique templates. SMS drops modestly (536 → 416) because Smishing-IMC25 is real user-reported data with high lexical diversity; only short (<30 chars) and duplicate messages are removed. Forum drops from 312 to 113 because many posts are short ("thanks, got it too") and fail the `len(text_clean) >= 30` filter, plus dedup and overlap with email/SMS.

Annotation: Google Gemini zero-shot, validated on 10% sample.  
Taxonomy (IFIT): 12 intents — 5 explicit requests + 7 implicit manipulations.

Method — four components

1. IFIT taxonomy. 12 intents, multi-label. Mean 2.97 intents per positive document.
2. Multi-Label SupCon. Positive pairs = intent overlap; cross-channel positives weighted  $\times(1+\alpha)$ ,  $\alpha = 0.5$ .
3. Kernel MMD. Multi-bandwidth RBF on CLS embedding,  $\sigma \in \{1, 2, 4, 8\}$ .
4. Post-hoc Channel Centering (PCC). Frozen encoder; subtract per-channel mean (estimated on train, applied to test). Linear, no retraining.



Joint objective:  $\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.3 \cdot \mathcal{L}_{\text{SupCon}} + 0.3 \cdot \mathcal{L}_{\text{MMD}}$

## Key results

| Metric                                        | Value                                                                                |
|-----------------------------------------------|--------------------------------------------------------------------------------------|
| mAP (macro, 9 intents)                        | 0.721                                                                                |
| Top-5 recall                                  | 0.931                                                                                |
| Email / SMS macro-F1                          | 0.451 / 0.452 (balanced — shortcut removed)                                          |
| Domain MI, raw CLS $\rightarrow$ centered CLS | 0.676 $\rightarrow$ 0.116 (−82.9%)                                                   |
| Domain MI, full model + PCC                   | 0.676 $\rightarrow$ 0.442                                                            |
| k-NN intent preserved under PCC               | $ \Delta  \approx 0.011$                                                             |
| Probability of Necessity, k = 5/10/20         | 0.207 / 0.294 / 0.320                                                                |
| Encoder comparison                            | UmBERTo 125M gives best mAP and Top-5; mDeBERTa has higher macro-F1 (0.501 vs 0.465) |

Ablation. Training-time objectives (SupCon, MMD) do not achieve PCC-level invariance. PCC is the decisive component — it works whether applied to the trained model or to the BCE-only baseline.

Figure A.  $\lambda$ -sweep on the Pareto frontier. Reducing channel information always costs F1;  $\lambda=0.3$  is the elbow.

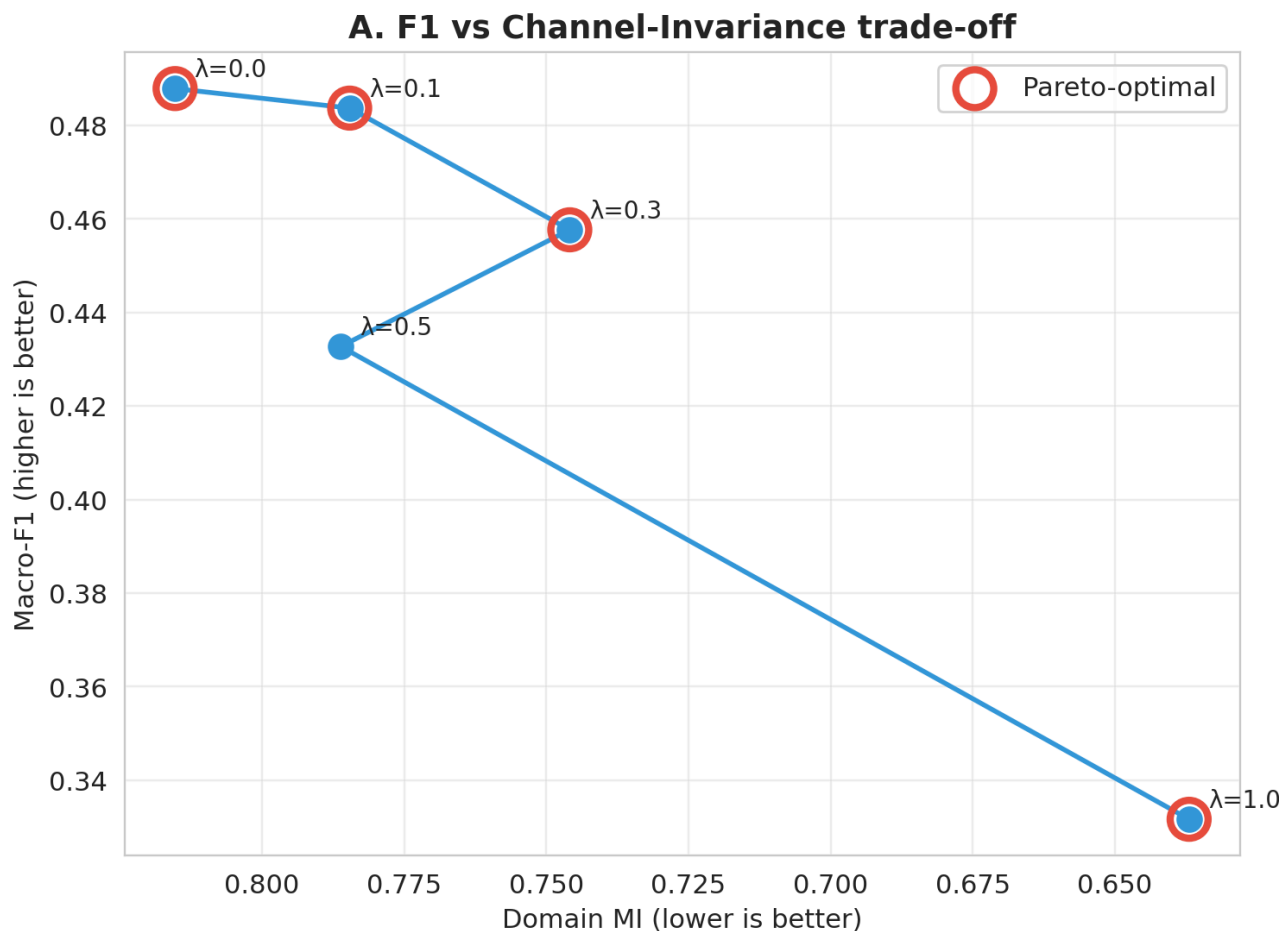


Figure B. PCC preserves intent signal:  $k$ -NN accuracy is essentially identical before (blue) and after (red) centering, while channel MI drops 83%.

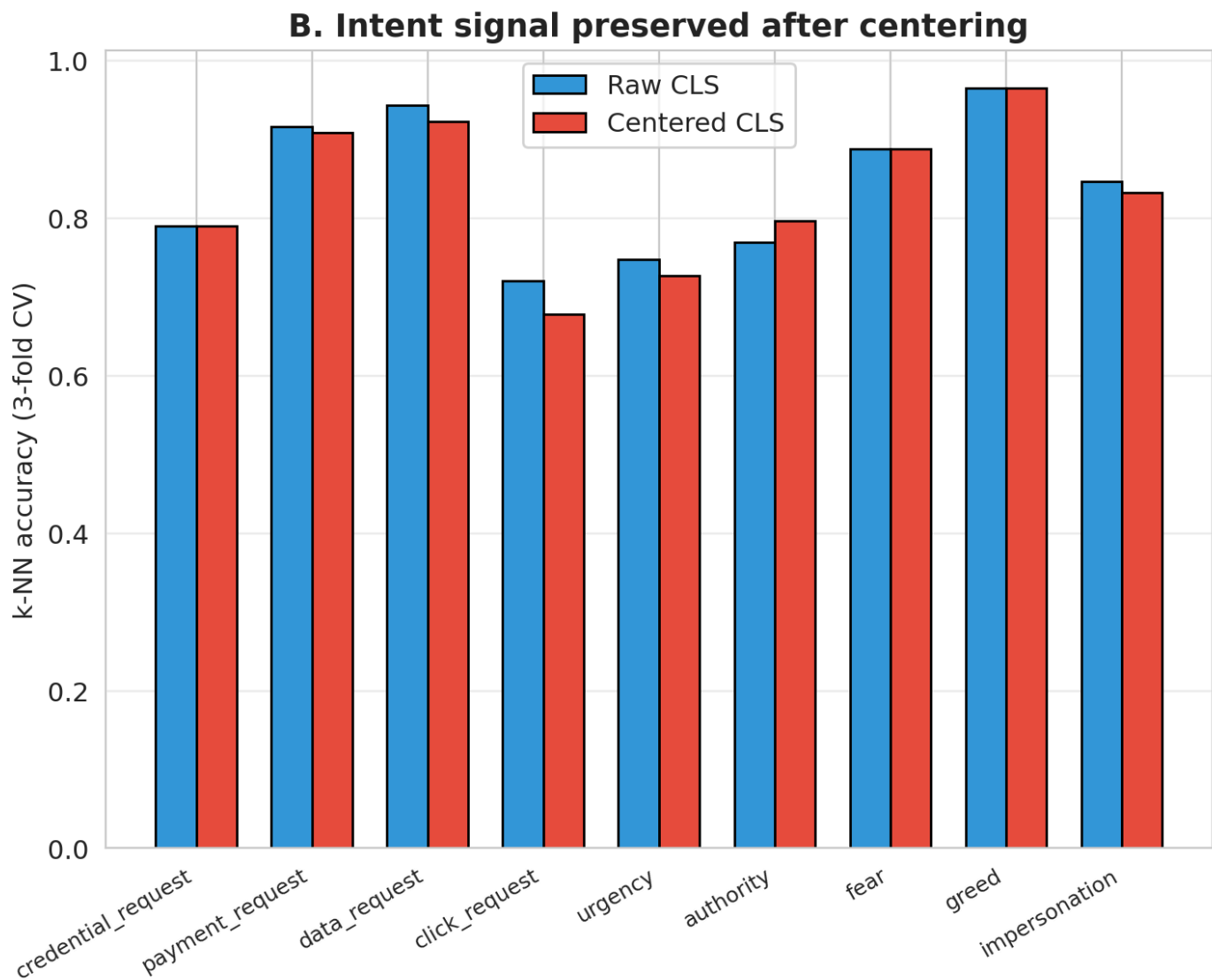
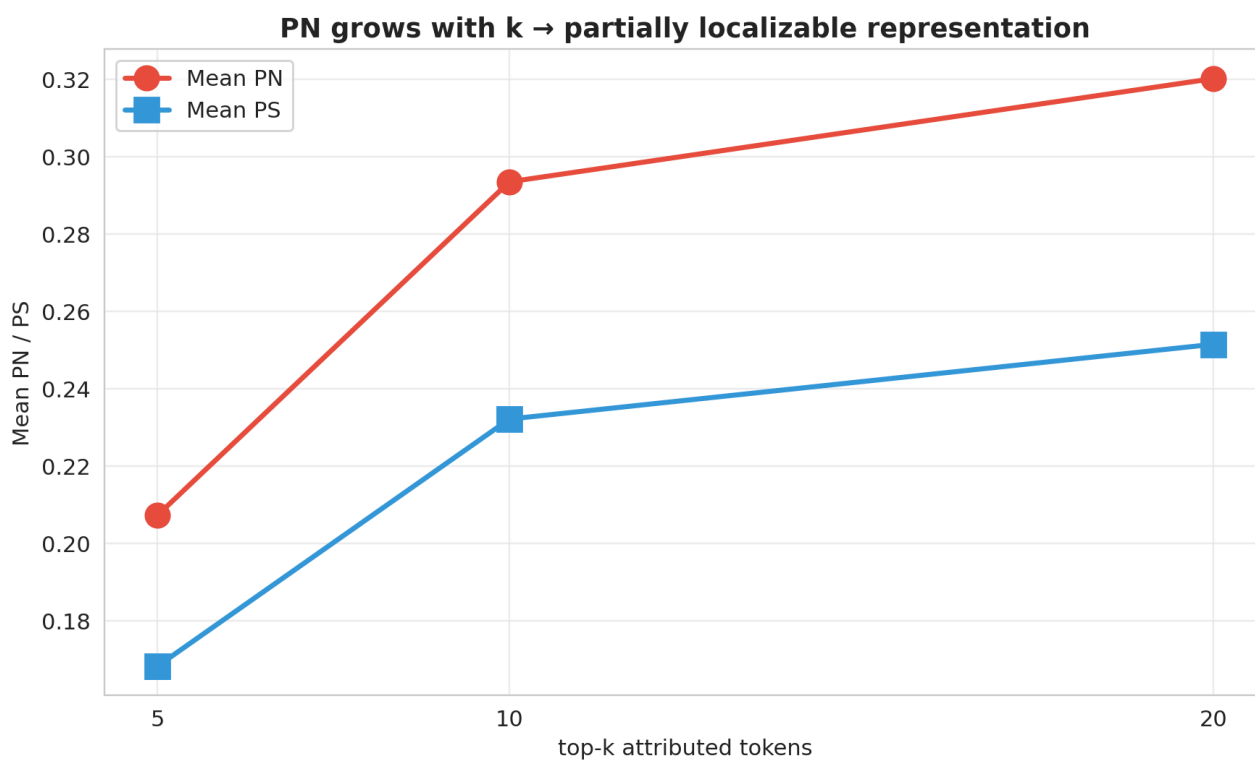


Figure C. PN grows sub-linearly with the number of attributed tokens — a signature of a partially localizable representation.



## Questions

1. **Use/mention confound.** Email and SMS carry use labels; forum carries mention. The model is trained jointly. We currently train jointly but report the use/mention gap. Is that acceptable, or would you recommend separating use-only training?
2. **LLM annotation.** 950 documents annotated zero-shot with Gemini, validated on 10%. No inter-annotator agreement. We plan 150–200 documents, 2 annotators, Cohen's  $\kappa$ . Would  $\kappa > 0.7$  be sufficient for IP&M?
3. **Novelty framing.** UmBERTo (125M) achieves the best Top-5 recall, outperforming larger multilingual encoders (278M). Is “channel shortcut diagnosis + architecture-agnostic post-hoc PCC” strong enough for IP&M, or should I reframe it as a methods paper on channel shortcuts?